Chess Robot

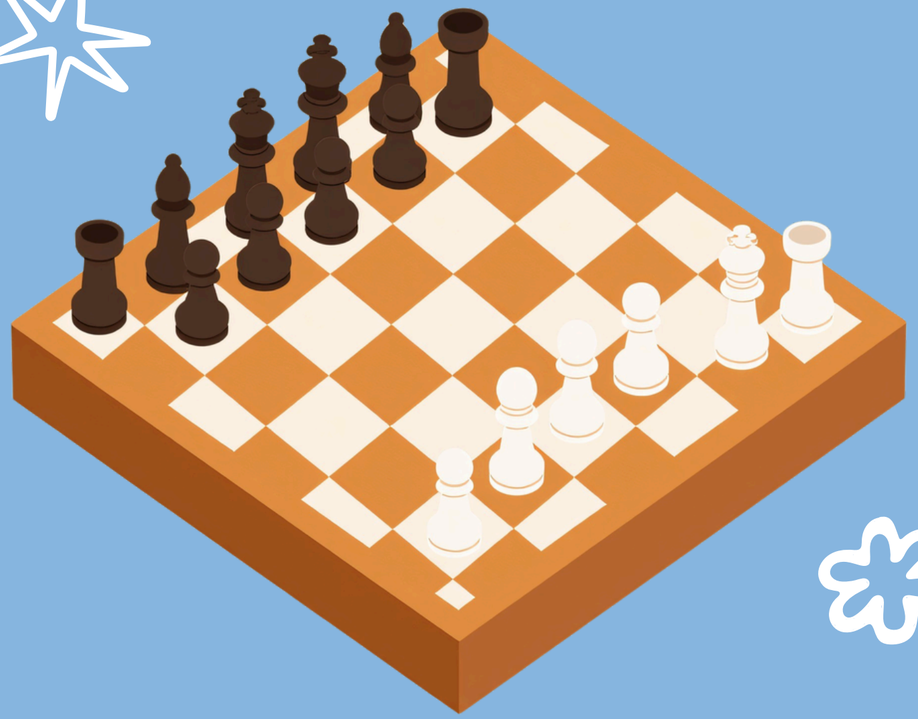
Project



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Topic



01 Idea

02 AI Concept

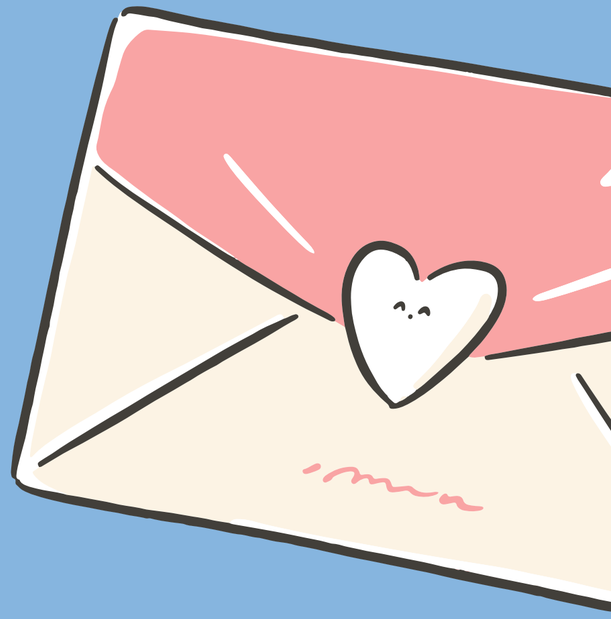
03 Object Detection Use

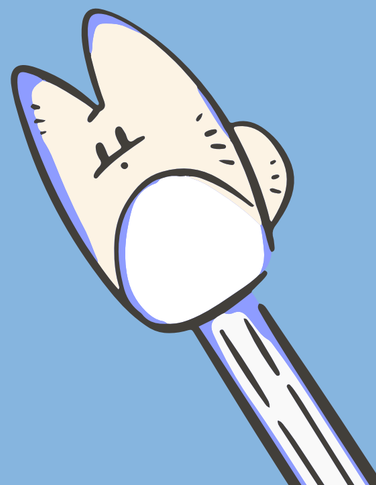
04 Expected Results

01

Idea



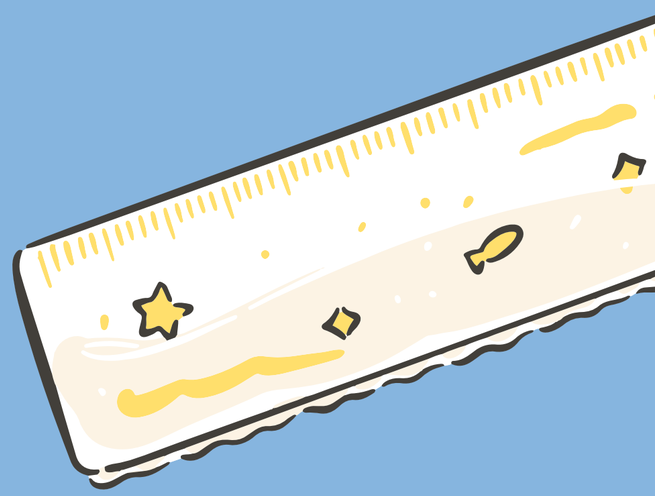




Idea

- We separate real life and simulation.
  - Real life is a real chessboard for players to interact.
  - Simulation is a display monitor for players to see events on the chessboard on both sides.
- Robot AI will update position on monitor every time. Show about what chesses have been moved.
  - If the player moves the chess pieces incorrectly, the Robotic AI arm (3D SIM) will fix it as same as you start your turn
- Notificate event on chessboard.



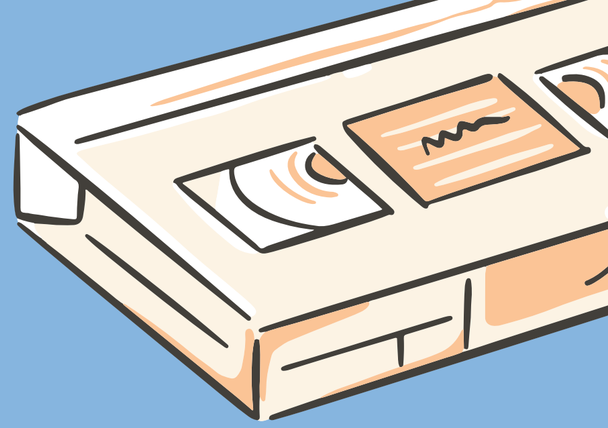




02

AI Concept









AI Concept

Hybrid "Digital Twin" Chess

- Play with real physical pieces while an AI camera tracks every move

Interactive Referee

- If a player makes an illegal move, the AI alerts them and the simulated arm visually resets the piece on screen.



Cost Effective Robotics

- A safe, zero risk simulation testbed for learning Inverse Kinematics (IK) without buying expensive physical robot arms.



03

Object Detection Use





Object Detection Use

- Robot AI will use a camera to detect objects on a chessboard.
- When an object is moved on a chessboard, the Robot AI will detect and analyze also check step on chessboard.

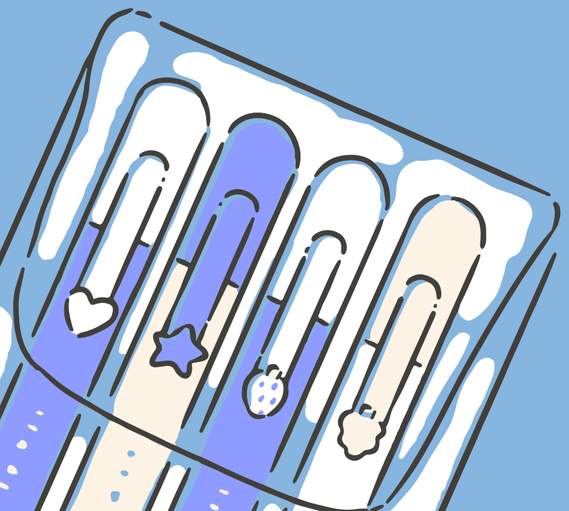
  - If the robot AI can detect something incorrect or interesting on board. Robot will notice what happen. For example, move wrong chess, checkmate and capture chess.

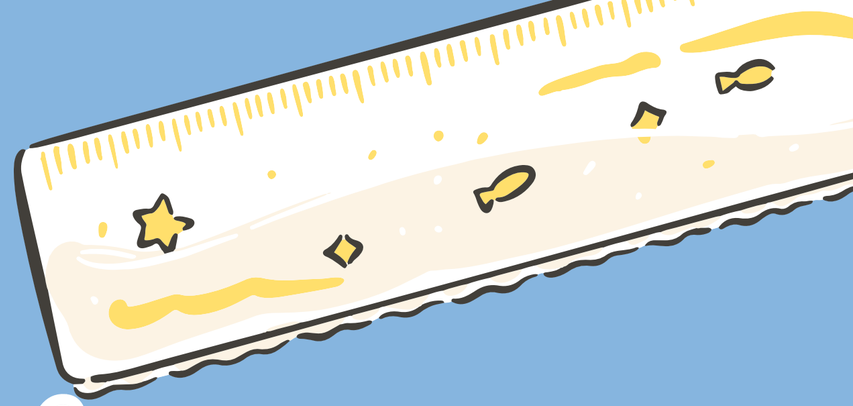
  - Notification will appear everytime when event occur same as example above.

- After finished analyze, Robot AI arm (3D SIM) will move chess pieces from their calculate. Repeat this until the end of the game.

04

Expected Results







Expected Results

- Robot AI will detect and analyze correctly.
- Robot AI can acknowledge what event happens on the chessboard.
- Robot AI can display monitor correctly and clearly on the screen.
- Robot AI can move chess pieces and fix chess correctly.

Expected Results

Hardware (Physical)

- Web Camera(1080p or higher)

  - Mounted above the board to capture real time video feed for piece detection.

- Physical Chessboard & Pieces

  - Used as the tangible, real world interface for the human player to interact with.

- Overhead Camera Mount / Tripod

  - To stably hold the web camera directly over the chessboard to prevent perspective distortion.

- Physical Chessboard & Pieces

  - Used by developers to run the 3-D simulation and engine, AI vision rendering, and backed logic simultaneously.

Expected Results

Software

- Python (Programming Language)

  - The primary backend language used to connect the computer vision, game logic and simulation.

- OpenCV (Vision Library)

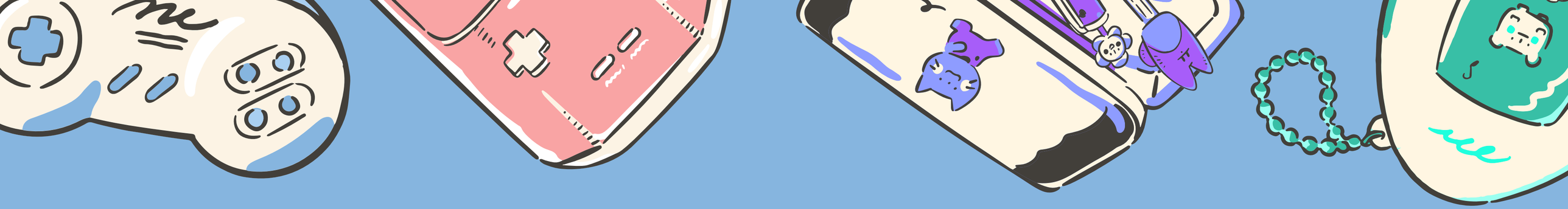
  - Used for image processing, tracking the physical chess pieces, and mapping pixels to logical board coordinates (A1-H8).with.

- Webots / Gazebo (Simulation Engine)

  - Used to create the 3D virtual environment and simulate the Robotics Arm's movements (Inverse kinematics).

- Stockfish (Chess AI Engine)

  - The brain of the system, used to validate illegal moves and calculate the computer's counter-moves.



Thank You

